EV Charging Project Alternatives — Two Concepts Beyond ChargeSight

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1. Recommendation

Choose Concept 1, ChargeProof. It addresses a documented verification gap, can be built and tested entirely in software, and its evaluation produces hard, checkable numbers. Concept 2 is a credible fallback; ChargeSight ranks third because most of its features already exist.

Rank	Concept	One line	Type of contribution
1	**ChargeProof**: statistical audit of real charger reliability	Certify how often public chargers actually work, using a fraction of the field visits	Established survey statistics applied to a documented, unsolved verification problem
2	**SiteSure**: payback-risk planner for small charger investors	See the odds a charger investment pays back before committing ₹30 lakh or more	Improvement and new combination of existing tools (financial model + site context + uncertainty)
3	ChargeSight (revised): explainable repair priority	Know which charger to fix first, and why	Improvement on features most CPMS vendors already sell

- **Why ChargeProof wins:** reliability bodies say there is no settled way to audit charger reliability, self-reported uptime is known to overstate reality, and India is now funding 72,300 public stations with subsidy tranches tied to performance benchmarks.
- **What it is not:** a new statistical method or a monitoring dashboard. The novelty is the audit design and the evidence it produces, which we must not overclaim.
- **Labels used in this doc:** *Verified* = stated on a cited page; *Vendor claim* = stated by the seller about itself; *Inference* = our reasoning; *Assumption* = a value we chose and must test.

2. Market scan: what we investigated and rejected

Seven directions were checked; five were rejected because established products or programmes already cover the essential capability, or because the data needed does not exist publicly.

Direction	What already exists (verified)	Verdict
Queue and reservation at busy chargers	ChargePoint Waitlist virtual queue; Tesla virtual queue pilot (Feb 2025); ChargeZone slot reservation; booking with auto-release in Unified Bharat e-Charge	**Rejected:** common feature
Interoperability adapter for small CPOs (UEI / Beckn)	UEI network reported 13,000+ sessions a day in Dec 2024 (Energetica); Pulse Energy says CPO setup takes under 30 minutes (vendor claim, Pulse)	**Rejected:** live network with onboarding providers
Load sharing at apartments and depots	Driivz time-of-use energy management; ChargeZone smart load management	**Rejected:** dynamic load management is standard
Truck charging siting	ICCT Scout Charge-Park built on GPS data from 200,000+ trucks (Sept 2026)	**Rejected:** data we cannot match
ML failure prediction	Claimed by Driivz and ChargerHelp; no public failure dataset for India	**Rejected:** breaks the no-ML-without-data rule
Verifying real charger reliability	Field programmes exist (UC Davis, ChargerAssess) but no public, cost-efficient audit design with confidence bounds was found	**Kept → Concept 1**
Investment risk for small charger investors	WRI FACt Excel IRR tool; Stable US site scoring; neither page mentions uncertainty modelling	**Kept → Concept 2** (as an improvement, not a novelty)

3. Concept 1: ChargeProof

ChargeProof plans and analyses field audits so a funder or operator can state, with a confidence interval, how often a network's chargers actually deliver a charge, using as few site visits as the required precision allows. The name has not been checked for trademarks.

A. Project identity

- **Pitch:** Certify how often public chargers really work, with a fraction of the field visits.
- **Simple explanation:** Operators report that chargers are "available". ChargeProof decides which chargers an auditor should physically test, records each test, and combines the test results with the operator's own status data to estimate the true charge-success rate, plus its margin of error.
- **Problem it solves:** Self-reported availability overstates what drivers experience, and there is no standard, affordable way to check it.
  - *Verified:* ChargerHelp's 2025 report found reported uptime of 98.7–99.9% but only about 71% of charging attempts succeeding, with over a third of failures on chargers that looked operational (Automotive World).
  - *Verified:* a Bay Area field study found 72.5% of fast-charger connectors functional against 95–98% reported (arXiv).
  - *Verified:* Atlas Public Policy says there is "no perfect technological solution" to audit charger reliability and leaves verification design to states (report).
  - *Verified (India):* PM E-DRIVE funds about 72,300 public stations from a ₹2,000 crore outlay, releasing subsidy in two tranches tied to compliance and performance benchmarks (Free Press Journal); IEEFA recommends linking support to verified uptime (IEEFA).
- **Who has the problem:** subsidy administrators and state nodal agencies (must judge performance), CPOs (want credible proof for site hosts, fleets and lenders), fleets and OEM partners (choose networks). Testing every charger is expensive; the UC Davis programme planned about 3,600 chargers of field tests (UC Davis).

B. Competitor and novelty validation

Player	What it does (per its page)	Field tests	Uses operator status data	Sample-size planning with confidence bounds	India
ChargerHelp ChargerAssess	Per-asset assessments of function, driver experience, maintenance readiness; 7,000+ assessments (vendor claim)	Yes	Not stated	Not described	Not stated
ChargerHelp reliability report with Paren data	Compares self-reported status with driver attempts; 19M+ data points	Partly	Yes	Not described	No
ChargeX Consortium	Defines KPIs: charge start success, session success, start-to-charge time	No	Guidance	No	No
UC Davis / CEC field testing	Research programme, standard test protocol, ~3,600 chargers	Yes	Not stated	Not described in the deck	No
NEVI reporting (summary)	97% uptime rule, operator self-reporting	No	Yes	No	No
IEEFA	Delhi sample study, ~84% non-functional (2024)	Yes	No	Not described	Yes
ChargeProof	Audit planner + test logger + estimator + certificate	Planned	Yes, as auxiliary data	**Core**	Yes

- **What already exists:** field testing, KPI definitions, reliability reports, and comparisons of reported vs actual status.
- **Proposed differentiator:** an audit-design engine that (1) computes how many tests are needed per stratum for a target margin, (2) uses the operator's status data for every charger to sharpen the estimate from a small tested sample, and (3) issues a certificate with the interval, method and seed.
- **What cannot be claimed as unique:** stratified sampling, confidence intervals and regression estimators are textbook survey statistics; reliability testing and reporting already exist. Large players may do this privately.

- **Classification:** a new application of established methods to a documented gap. Not a new algorithm; not patentable on the evidence we have.

C. Working prototype (8–9 weeks)

Screens

1. **Network registry:** map and table of a simulated network of 500 chargers from 5 operators, with type (DC/AC), age band and reported status.
2. **Audit planner:** enter target margin (e.g. ±5 points), confidence (95%), visit budget and cost per visit; get visits per stratum and a visit list.
3. **Field test logger:** mobile web form per visit: arrived, screen working, payment accepted, charge started, energy after 2 minutes, failure code. In the prototype a simulated auditor fills it.
4. **Certificate:** per operator, reported availability vs verified session success with 95% interval, sample size, method, seed.
5. **Experiment lab:** runs the evaluation in Section D and plots error, interval coverage and visits needed.

Core logic

```
n_0 = \frac{z^2\,p(1-p)}{e^2}, \qquad n = \frac{n_0}{1 + (n_0 - 1)/N}
```

Worst case p = 0.5 at 95% confidence: ±5 points needs 218 tests for N = 500, ±8 points needs 116. Strata get Neyman allocation. The assisted estimator is the two-phase regression estimator, using reported status (known for all chargers) as the auxiliary variable; its variance falls roughly by a factor of $(1 - \rho^2)$, where ρ is the correlation between reported status and test outcome. Intervals use Wilson for single proportions and the stratified variance formula otherwise.

Data generation: each simulated charger has a hidden true success probability set by operator quality, age and type, and a reported status that can be wrong ("ghost available") at a configurable rate. Because truth is known in simulation, every estimate can be scored. All rates are *assumptions*, swept in the experiment.

Stack: Python, FastAPI, NumPy/SciPy, SQLite or PostgreSQL; React or plain HTML with Leaflet and OpenStreetMap tiles.

Example (illustrative numbers, not a result): Operator A, 120 chargers, reports 97% available. The planner asks for 40 tests; 29 succeed. ChargeProof reports session success of 72.5% with a Wilson 95% interval of about 57–84% before using the status data, then a narrower assisted interval.

Faculty demo: "Operator claims 97%. We test 40 of 120 chargers. Here is the verified rate, the margin of error, and why we did not need to test all 120." Then move the ghost-rate slider and watch naive uptime stay high while the verified estimate tracks the hidden truth.

Without hardware or private data: the network, statuses and test outcomes are generated; the method is identical when real test results are entered.

MVP: planner, simulated network, logger, estimators, certificate, experiment lab.
Excluded: real operator data feeds, scraping third-party apps, payment for test sessions, mobile app store release, ML.

D. How to prove it works

- **What is tested:** whether the stratified, status-assisted estimator hits a target margin with fewer tests than simple random sampling, while keeping honest interval coverage.
- **Baselines:** B0 = operator-reported availability (no tests); B1 = simple random sample, plain proportion.
- **Design:** 1,000 Monte Carlo audits per scenario; scenarios sweep true success (60–95%), ghost rate (0–20%), and how different the strata are. Same seeds for every method.
- **Metrics:** absolute error vs hidden truth; 95% interval coverage (should be near 95%); interval width; tests needed to reach ±5 points; audit cost at an assumed cost per visit.
- **Success:** across most scenarios the proposed method reaches ±5 points with fewer tests than B1, and coverage stays at or above 93%. **Failure:** coverage drops well below 95%, or no saving appears. Theory predicts little saving when reported status barely correlates with real success; the report must show where that happens.
- **Not claimable without evidence:** any saving for real networks, any real operator's success rate, or that simulated ghost rates match India.

E. Entrepreneurial opportunity

- **Initial customers:** subsidy administrators and state nodal agencies under PM E-DRIVE (the performance-linked tranche is the hook); second, CPOs wanting third-party proof of reliability.
- **Value proposition:** a defensible, auditable reliability figure at a known cost, instead of trusting self-reports or testing everything.
- **Alternatives:** trusting operator dashboards; one-off studies (IEEFA-style); full field assessments (ChargerAssess-style).

- **Revenue options:** audit-as-a-service fee per operator per period; software licence for agencies running their own auditors; certificate badge for CPOs. No prices set; must come from interviews.
- **Why pay:** funds released on unverified performance carry public-money risk; a CPO with a credible certificate may win site hosts and fleet contracts (*inference*).
- **Validate in interviews:** whether agencies are required or willing to verify; who pays; acceptable margin; real cost per visit; whether operators will share status logs.
- **Risks:** technical (weak correlation reduces savings); commercial (slow government buying); ethical (a wrong low estimate harms an operator's reputation, so publish method, interval and a dispute process); legal (no scraping of apps without permission).

F. PME deliverables

PME item	How ChargeProof supports it
Problem, objectives	Reported vs real reliability gap; objective = target margin at minimum visits
Scope, feasibility	Software-only; feasibility shown by the simulation
WBS	Data generator → planner → estimators → logger → certificate → experiment → business case
Network, critical path, Gantt	Estimator and experiment sit on the critical path; UI runs in parallel
Resources, budget	Four students, laptops, free hosting; field-visit cost as an explicit assumption
Risk register	Correlation, coverage, adoption, ethical risks above
Business Model Canvas	Agencies and CPOs; audit service and licence
Prototype	Planner + certificate + experiment lab
Market, competitors	Table in B
Outcomes, pitch	Measured visit savings and coverage from D

4. Concept 2: SiteSure

SiteSure shows a small investor the probability that a proposed charging station pays back within their horizon, which assumptions drive that probability, and how two or three candidate sites compare. It improves on existing tools; it is not a new category. The name

has not been checked for trademarks.

A. Project identity

- **Pitch:** Before you put ₹30 lakh into a charger, see the odds it pays back.
- **Simple explanation:** The investor picks a site on a map and enters ranges, not single guesses, for utilisation, tariff, price and costs. SiteSure runs thousands of scenarios and reports the chance of payback, the downside case, and the break-even utilisation.
- **Problem it solves:** returns depend mostly on utilisation, which is uncertain before opening, yet common tools give one number.
  - *Verified (vendor page):* Statiq's franchise model starts from ₹30 lakh for a 60 kW DC station; Statiq says returns depend on location, capacity and utilisation and warns a low-demand site slows utilisation (Statiq).
  - *Verified:* WRI India's FACt is an Excel tool reporting IRR; its page mentions neither per-site demand nor uncertainty (WRI).
  - *Verified:* only about 71% of FAME-II-approved stations were operational in March 2026 (Bolt.Earth); *inference:* poor siting and weak economics are plausible contributors, not proven ones.
- **Who has the problem:** petrol-pump owners, mall and hotel owners, landowners and small businesses considering franchise or owned stations; small CPOs choosing their next site; lenders assessing such loans.

B. Competitor and novelty validation

Player	What it does (per its page)	India	Site context	Financial model	Uncertainty / probability of payback
WRI FACt	Excel financial evaluation, IRR vs target; Creative Commons	Yes	Not mentioned	Yes	Not mentioned
Stable	ML site score trained on millions of real sessions (vendor claim)	No (US)	Yes	Not described	Not described
ICCT Scout Charge-Park	Truck charging hotspots + grid overlay	Yes	Yes (trucks)	No	No
Statiq FOCO	Includes "site demand validation" for franchisees	Yes	Yes (internal)	Not public	Not public
SiteSure	Site context + cash-flow model + Monte Carlo	Yes	Public data	Yes	**Core**

- **What already exists:** charging financial models (FACt), data-driven site scoring (Stable, ICCT), CPO-run site validation.
- **Proposed differentiator:** decision under uncertainty for the *investor*, independent of the CPO selling the franchise: probability of payback, P10/P50/P90 NPV, break-even utilisation, and the assumptions that move the answer most.
- **What cannot be claimed as unique:** Monte Carlo cash-flow analysis is standard in project finance; site scoring exists; CPOs may already do this internally.
- **Classification:** improvement and new combination of existing features. Weaker novelty than ChargeProof.

C. Working prototype (8–9 weeks)

Screens

1. **Site picker:** Leaflet map on OpenStreetMap tiles; click a candidate site.
2. **Site context:** counts of nearby points of interest from OpenStreetMap and existing chargers within a radius from the Open Charge Map API, each shown with its source.
3. **Assumptions editor:** capex, charger size, electricity tariff, selling price, fixed costs, revenue share, ramp-up, and utilisation as a low / likely / high range.
4. **Results:** payback-time distribution, probability of payback within the investor's horizon, P10/P50/P90 NPV, break-even utilisation, tornado chart of the biggest drivers.
5. **Compare sites:** two or three sites side by side.

Core logic

$$E_t = P \times 8760 \times u_t \times \eta, \qquad CF_t = E_t\,(s - c) - F_t$$

E is energy sold per year, P charger power, u utilisation, η efficiency, s selling price per kWh, c energy cost per kWh, F fixed costs and revenue share. Draw u, s, c and F from the entered ranges 10,000 times; compute NPV and payback per draw. A deterministic mode uses the likely values only, so results can be checked against a hand or spreadsheet calculation.

Data: OpenStreetMap (ODbL) for context; Open Charge Map for existing chargers, which licenses each record separately, so attribution must follow each record (OSM wiki). Utilisation stays an **assumption range** the user enters; context data informs it but never silently sets it.

Stack: Python, FastAPI, NumPy, React or plain HTML with Leaflet and a charting library.

Example input (assumptions): 60 kW DC, capex ₹30 lakh, utilisation low 5% / likely 10% / high 18%, tariff and price from the current state order, 7-year horizon. **Output shape:** "Chance of payback within 5 years: X%; break-even utilisation: Y%; biggest driver:

utilisation." No numbers are claimed here.

Faculty demo: two Bengaluru sites with the same capex. Their single-number IRR looks similar; the distributions show one has a much higher chance of losing money.

Excluded: ML demand forecasting, grid hosting capacity, real CPO utilisation data, loan origination.

D. How to prove it works

- **What is tested:** (1) correctness of the model; (2) whether deciding on probability of payback leads to fewer loss-making investments than deciding on a single-point IRR, when utilisation is uncertain.
- **Correctness test:** deterministic mode must match an independent spreadsheet for 10 fixed input sets to the rupee; Monte Carlo results must stabilise as draws increase.
- **Decision test (synthetic):** generate 1,000 hypothetical sites, each with a hidden true utilisation; the investor sees only a noisy range. Rule A invests if point IRR ≥ target; Rule B invests if P(NPV > 0) ≥ 70%. Score the share of investments that lose money and the total realised NPV. Sweep noise level and the 70% threshold.
- **Success:** Rule B cuts loss-making investments without giving up most of Rule A's total NPV across most settings. **Failure:** no difference, or gains only at one tuned threshold.
- **Usability check:** five participants compare three sites with SiteSure and with a spreadsheet; record time and errors.
- **Not claimable:** forecast accuracy for real sites, real investor savings, or that our utilisation ranges reflect Indian reality.

E. Entrepreneurial opportunity

- **Initial customers:** prospective franchise investors and small CPOs choosing sites; later, banks and NBFCs assessing charger loans.
- **Value proposition:** an independent, transparent second opinion before a large, hard-to-reverse investment.
- **Alternatives:** the CPO's own pitch and site validation, consultants' feasibility studies, FACt, gut feel.
- **Revenue options:** per-site report fee; subscription for brokers and consultants; white-label for lenders. No prices set.
- **Why pay:** *inference* only: a small fee compared with capex in the lakhs, if investors trust the method.
- **Validate in interviews:** do investors distrust CPO projections; who would pay; what horizon and threshold they use; whether any CPO will share utilisation benchmarks.

- **Risks:** technical (garbage in, garbage out on utilisation); commercial (CPOs may see it as a threat; trust); ethical and legal (false confidence, liability for losses: label as decision support, not financial advice; respect per-record data licences).

F. PME deliverables

PME item	How SiteSure supports it
Problem, objectives	Uncertain returns on large investments; objective = quantify payback risk
Scope, feasibility	Public data and user inputs only
WBS	Data adapters → cash-flow model → Monte Carlo → UI → experiments → business case
Network, critical path, Gantt	Model and experiment on the critical path; map UI in parallel
Resources, budget	Four students, free APIs within limits, free hosting
Risk register	Utilisation assumptions, trust, liability, licensing
Business Model Canvas	Investors and lenders; report fees and licences
Prototype	Map + assumptions + distribution results
Market, competitors	Table in B
Outcomes, pitch	Decision-test results and usability check from D

5. Final comparison

ChargeProof leads on differentiation, evaluation strength and viva defence; SiteSure has the most visual demo but the weakest evidence; ChargeSight has the highest competitor overlap. Ratings are our judgement from the research above.

Criterion	ChargeProof	SiteSure	ChargeSight (revised)
Novelty and differentiation	Medium-high: documented verification gap	Medium-low: improvement on FACt and site scoring	Low-medium: explanation layer on common features
Competitor overlap	Low-medium (field assessments exist)	Medium (financial tools, site scoring)	High (Driivz, Monta, Ampcontrol, others)

Criterion	ChargeProof	SiteSure	ChargeSight (revised)
Technical difficulty	Medium: survey statistics, simulation	Medium: cash-flow model, Monte Carlo, maps	Medium: OCPP, queue simulation
Prototype feasibility	High	High	High
Data-access risk	Low for prototype; real audits need field access	Medium: utilisation stays an assumption	Low for prototype
Demonstration strength	High: claimed vs verified, with ghost-rate slider	High: maps and distributions	Medium: ranked ticket list
Entrepreneurial potential	Medium-high: policy hook (PM E-DRIVE tranches), but slow government buyers	Medium: clear buyer, trust and liability issues	Low-medium: crowded market
Risk of being challenged in viva	Low-medium if statistics are understood	Medium-high: "where does utilisation come from?"	Medium-high: "Driivz already does this"

Ranking: 1. ChargeProof, 2. SiteSure, 3. ChargeSight.

For a four-person team with 8–9 weeks, choose **ChargeProof**. Its core is small and testable (sample-size planner, estimators, simulation), its results are objective (error, coverage, tests saved against a known truth), and its problem is backed by independent evidence rather than vendor claims. The one skill to secure early is the statistics: assign one person to stratified sampling and the regression estimator in week 1, and validate the estimator code against textbook examples before building screens.

Sources

- ChargerHelp 2025 reliability report (Automotive World)
- Reported vs actual uptime, ChargerHelp and Paren data (Electrek)
- ChargerHelp ChargerAssess
- Rempel et al., reliability of DC fast chargers (arXiv)
- Atlas Public Policy, evaluating charger reliability
- UC Davis EV charger field testing workshop

- ChargeX customer-focused KPIs (INL)
- NEVI final rule summary (Zeta)
- PM E-DRIVE charging guidelines (Free Press Journal) · Powerline
- IEEFA, EV charging experience in India (Sept 2026)
- Bolt.Earth, 71% operational · Unified Bharat e-Charge explainer
- WRI India FACt tool
- Stable site selection (Business Wire)
- ICCT Scout Charge-Park (Autocar Pro)
- Statiq franchise guide
- Open Charge Map licensing (OSM wiki)
- ChargePoint Waitlist · Tesla virtual queue (Notebookcheck) · ChargeZone wait time
- UEI daily sessions (Energetica) · Pulse Energy on UEI
- Driivz operational excellence